

Date:30-06-2026

Lab Activity 1 : Problem Identification, Analysis, and Solution Generation CO1

Objective

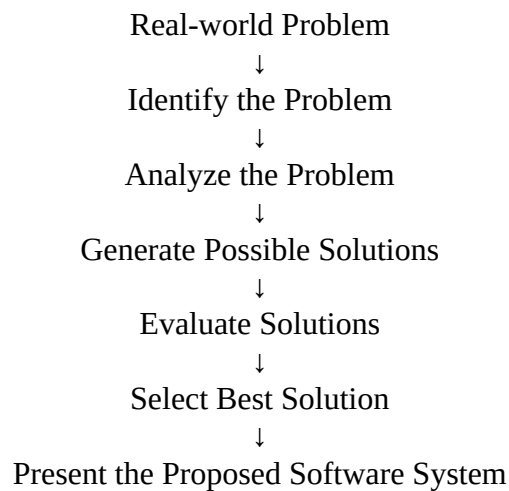
To enable students to identify a real-world software problem, analyze its causes and requirements, and propose a suitable software solution using Software Engineering principles.

Learning Outcomes

After completing this activity, students will be able to:

- Identify real-world software problems.
- Analyze stakeholders and requirements.
- Differentiate between functional and non-functional requirements.
- Generate multiple solution alternatives.
- Select the best software solution with justification.

Activity Workflow



Part 1

Complete the table.

Question	Answer
Problem Title	Real-Time Ambulance Route Blockage Prediction System
Who faces this problem?	Ambulance Drivers, Paramedics, Hospital Emergency Departments, Traffic Control Authorities, Patients
Why does it exist?	No system exists to predict and alert ambulances about sudden traffic blockages (accidents, congestion, road closures) on their route in real time.
Current solution	Manual radio communication with dispatch, driver's local road knowledge, and generic GPS navigation apps not designed for emergency vehicles.
Limitations	Delayed response to sudden traffic changes, no real-time rerouting, increased risk to patients due to delays, no integration with traffic signal systems.

Example

Field	Answer
Problem Title	Manual Attendance System
Who faces the problem?	Faculty, Students, Department Administration
Why does the problem exist?	Attendance is recorded manually on paper or spreadsheets, making the process slow, error-prone, and difficult to manage.
Current Solution	Faculty record attendance in paper registers or Excel sheets. Attendance reports are prepared manually at the end of each week or month.
Limitations of the Current Solution	Time-consuming process, Proxy attendance is difficult to detect, Paper records can be lost or damaged, Manual calculation errors, Difficult to generate attendance reports, No real-time attendance monitoring, Requires significant manual effort, Difficult to track attendance history, Limited accessibility to records, High maintenance of physical registers

Part 2: Problem Analysis

Stakeholders: Ambulance Drivers, Paramedics, Hospital Emergency Staff, Traffic Police, Patients, Emergency Dispatch Centers

Identify Stakeholders

Constraints: Real-time data availability, GPS accuracy, network connectivity, integration with traffic signal systems, limited budget, data privacy.

Constraints

Risks: Inaccurate traffic predictions, GPS/network failure, delayed data updates, false alerts, system downtime during emergencies.

Risks

Part 3: Root Cause Analysis

Students identify why the problem exists.

Answer: There is no centralized, real-time traffic monitoring system integrated with ambulance routing. Traffic data is scattered across multiple sources and is not predictive, causing delayed rerouting when sudden blockages occur.

Part 4: Generate Solutions

Each team generates at least three possible software solutions.

Solution 1 Desktop-based Traffic Monitoring Dashboard for Dispatch Centers

Solution 2 Mobile App for Real-Time Ambulance Route Blockage Prediction

Solution 3 Cloud-based Smart Ambulance Routing and Blockage Prediction System

Part 5: Evaluate Solutions

Students compare solutions.

Low, Medium, High, Easy, Criteria	Desktop	Mobile	Cloud
Cost	Low	Low	Medium
Security	Medium	High	High

Accessibility	Low	High	High
Maintenance	Medium	High	High
Scalability	Low	High	High

Choose the best solution and justify it.

Best Solution: Cloud-based Smart Ambulance Routing and Blockage Prediction System because it integrates real-time data from multiple sources, is scalable, accessible from any device, easy to maintain, and provides accurate live blockage predictions without heavy on-site infrastructure.

FACULTY

Criteria	Marks
Problem Identification	3
Requirement Analysis	4
Root Cause Analysis	4
Solution Generation	4
Justification	3
Presentation	2
Total	10